

NovaDiff — Target codebase

Baseline: 9aadd5eb5604

Target: NovaDiff

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Diagrams use Mermaid heuristics (imports, cross-file call guesses, type sketches). They are **not** compiler-grade call graphs. The module graph is file-to-file (JS/TS relative imports). The call graph links files when a callee name maps to exactly one definition in the repo (regex across many languages).

AI target codebase narrative

Repository overview

NovaDiff is a cross-platform desktop tool that visualises and analyses changes in a codebase. Key parts of the repository are:

Layer	Purpose	Main entry points
Electron	UI and orchestration	electron/main.cjs , electron/preload.cjs
Rust CLI	Diff engine and Git helpers	cli/src/main.rs
Graph-Core	Graph building, LLM analysis, embeddings	packages/graph-core/src
LLM integration	Prompt generation & response parsing	electron/llm.cjs
Git / GitHub	Repository discovery, PR handling	electron/git-service.cjs , electron/github-service.cjs
Security scan	Vulnerability detection	electron/security-scan-runner.cjs
Workspace store/history	Snapshotting & navigation	electron/workspace-store.cjs , electron/workspace-history.cjs

The repository is organised as a monorepo under packages/ (primarily graph-core) and an Electron bundle that consumes those packages.

Architectural layout



- The Electron **main** process exposes APIs to the renderer via the **preload** script.
- Diff requests are forwarded to the **Rust CLI** (cli/src/main.rs).
- The **Graph-Core** library is imported by the Electron renderer and by the Rust CLI (via a Node bridge).
- LLM prompts are generated in electron/llm.cjs and sent to external providers (Ollama, LM Studio).

- GitHub authentication flows use the `gh` CLI (`electron/github-auth-flow.cjs`).
- Security scans are launched from the main process and parse audit outputs from `npm`, `pip`, and `cargo`.

Key subsystems

- **Electron UI & orchestration**
 - `electron/main.cjs` – window creation, menu, IPC handlers.
 - `electron/preload.cjs` – safe API surface for the renderer.
 - `electron/diff-excerpt.cjs` – builds diff snippets for the UI.
 - `electron/export-snapshot.cjs` – zips workspace snapshots.
 - `electron/prefetch-summaries.cjs` – background worker that pre-loads LLM summaries.
- **Rust CLI** (`cli/src/main.rs`)
 - `ChangeKind` enum – classifies file changes.
 - `walk_path_allowed`, `filter_changes_for_prefetch_gitignore` – path filtering logic.
 - `try_build_gitignore_for_root` – generates a `.gitignore` for the workspace root.
 - Exposed via `electron/compare-runner.cjs` (`runCompareEngine`, `parseEngineStdout`).
- **Graph-Core** (`packages/graph-core/src`)
 - **Analyzer** – builds a graph of files, functions, and calls.
 - `GraphBuilder` (`src/analyzer/graph-builder.ts`) – core graph construction.
 - `normalize-graph.ts` – node/edge normalisation.
 - `change-classifier.ts` – determines update decisions.
 - `layer-detector.ts` – assigns LLM-derived layers.
 - **LLM integration** – prompts and response parsing (`llm-analyzer.ts`).
 - **Embedding search** – semantic search over file embeddings (`embedding-search.ts`).
 - Tests live under `packages/graph-core/src/__tests__`.
- **Git & GitHub**
 - `electron/git-service.cjs` – thin wrapper around `git` CLI.
 - `electron/github-service.cjs` – uses `gh` CLI for PRs, repos, auth.
 - `electron/gh-install.cjs` / `electron/gh-path.cjs` – install & path resolution helpers.
- **Security scanning** (`electron/security-scan-runner.cjs`) – orchestrates `npm audit`, `pip audit`, `cargo audit`, parses advisories, maps severity to confidence, and emits markers.
- **Workspace persistence** (`electron/workspace-store.cjs`, `electron/workspace-history.cjs`) – JSON-based session store and commit-based snapshot indexing.

Dependency signals

Source	Target	Edge type	Notes
<code>electron/main.cjs</code>	<code>electron/preload.cjs</code>	IPC	Exposes APIs
<code>electron/main.cjs</code>	<code>electron/git-service.cjs</code>	import	Git helper
<code>electron/main.cjs</code>	<code>electron/github-service.cjs</code>	import	GitHub helper
<code>electron/main.cjs</code>	<code>electron/security-scan-runner.cjs</code>	import	Security scans
<code>electron/main.cjs</code>	<code>electron/knowledge-graph-runner.cjs</code>	import	Graph building
<code>electron/main.cjs</code>	<code>electron/llm.cjs</code>	import	LLM prompts
<code>electron/knowledge-graph-runner.cjs</code>	<code>packages/graph-core/src</code>	import	Graph-Core library
<code>electron/llm.cjs</code>	<code>electron/llm.cjs</code>	internal	LLM config
<code>electron/compare-runner.cjs</code>	<code>cli/src/main.rs</code>	spawn	Rust diff engine
<code>electron/compare-runner.cjs</code>	<code>electron/git-service.cjs</code>	import	Repo context
<code>electron/compare-runner.cjs</code>	<code>electron/github-service.cjs</code>	import	PR context
<code>packages/graph-core/src</code>	<code>packages/graph-core/src/analyzer</code>	import	Analyzer modules
<code>packages/graph-core/src</code>	<code>packages/graph-core/src/embedding-search</code>	import	Embedding engine
<code>packages/graph-core/src</code>	<code>packages/graph-core/src/language-lesson</code>	import	Language detection
<code>packages/graph-core/src</code>	<code>packages/graph-core/src/layer-detector</code>	import	Layer assignment

Operational considerations

- **Build**
 - Rust CLI: `cargo build --release` → `cli/target/release/cli`.
 - Electron: `npm run build` (uses `electron-builder`).
 - Graph-Core: TypeScript compiled via `tsc` (part of Electron build).
- **Runtime**
 - Requires `git`; `gh` (GitHub CLI) is optional.
 - LLM provider: Ollama or LM Studio; configured in `electron/llm.cjs`.
 - Security scans run asynchronously; results are cached in workspace snapshots.
- **Packaging**
 - `electron-builder` produces platform-specific installers.
 - The Rust binary is bundled under `resources/cli`.
- **Testing**
 - Unit tests: `packages/graph-core/src/__tests__`.
 - Integration tests: `tests/` (not fully scanned).
 - LLM tests use mock responses (`packages/graph-core/src/__tests__/...`).
- **Performance**
 - Diff engine is CPU-bound; runs in a child process.
 - Embedding search uses `SemanticSearchEngine` (cosine similarity).
 - Prefetch summaries run in a worker thread to avoid UI blocking.

Documentation gaps

- Electron renderer API surface – no explicit README or API surface description.
- CLI usage guide – `cli/src/main.rs` is documented in code, but no high-level CLI guide.
- Graph-Core public API – no generated docs or example usage.
- LLM configuration – environment variables or config files are not described.
- Security scan output – format and interpretation are not documented.
- Workspace store schema – JSON structure is inferred but not formally described.
- Testing strategy – coverage metrics and test harness details are missing.

Suggested onboarding and verification

1. Clone & install

```
git clone https://github.com/yourorg/NovaDiff
cd NovaDiff
npm ci
cargo fetch
```

2. Build Rust CLI

```
cargo build --release
```

3. Run Electron in dev mode

```
npm run dev
```

4. Verify core functionality

- Open a sample repo in the UI.
- Trigger a diff and inspect the graph.

- Check that LLM prompts appear in the console.

5. Run unit tests

```
npm test
```

6. Run integration tests (if available)

```
npm run test:integration
```

7. Validate security scans

```
npm run security:scan
```

8. Build installers

```
npm run build
```

9. Review generated docs (if any)

- Check `packages/graph-core/docs` or similar.

10. Cross-check API surface

- Inspect `electron/preload.cjs` for exposed functions.
- Verify that renderer code imports these correctly.

Target codebase metrics

Target codebase metrics

Repository: NovaDiff

Path: /Users/iamgroot/Documents/GitHub/NovaDiff

Metric	Value
Indexed files	484
Symbol-bearing files	80
Relative import edges	45
Cross-file call edges (heuristic)	220

Top extensions

Extension Count

.ts	226
.tsx	85
.pak	58
.cjs	28
(no ext)	20
.json	14
.plist	9
.css	6
.dylib	4
.mjs	4
.svg	4
.md	3
.yaml	3
.blockmap	2
.html	2
.icns	2
.asar	1
.bin	1
.dat	1
.dmg	1

Top directories

Directory	Files
packages	237
release	103
src	84
electron	28
.	15
tests	7
.novadiff-graph	4
cli	3
public	2
scripts	1

Detected project stacks

- **node:** package.json

Sample files

- .DS_Store
- .gitignore
- .novadiff-graph/config.json
- .novadiff-graph/diff-overlay.json
- .novadiff-graph/knowledge-graph.json
- .novadiff-graph/meta.json
- CONTRIBUTIONS.md
- LICENSE
- README.md
- THIRD_PARTY_NOTICES.md
- cli/Cargo.lock
- cli/Cargo.toml
- cli/src/main.rs
- electron/code-city-model.cjs
- electron/compare-runner.cjs
- electron/diff-excerpt.cjs
- electron/export-snapshot.cjs
- electron/file-summary-export.cjs
- electron/gh-install.cjs
- electron/gh-path.cjs
- electron/git-blame.cjs
- electron/git-commit.cjs
- electron/git-publish.cjs
- electron/git-service.cjs
- electron/github-auth-flow.cjs
- electron/github-commit-context.cjs
- electron/github-service.cjs
- electron/graph/extract-structure.mjs
- electron/knowledge-graph-runner.cjs
- electron/llm.cjs
- electron/main.cjs
- electron/novadiff-docs-html.cjs
- electron/novadiff-docs-pdf.cjs
- electron/novadiff-docs-writer.cjs
- electron/prefetch-summaries.cjs
- electron/preload.cjs
- electron/repo-discovery.cjs
- electron/security-scan-runner.cjs
- electron/workspace-files.cjs
- electron/workspace-history.cjs
- electron/workspace-store.cjs
- index.html
- nova-diff-desktop.jpg
- nova-diff-icon.png
- package-lock.json
- package.json
- packages/graph-core/package.json
- packages/graph-core/scripts/generate-alphabetical-catalog.mjs
- packages/graph-core/scripts/language-names.txt
- packages/graph-core/src/__tests__/change-classifier.test.ts
- packages/graph-core/src/__tests__/domain-normalize.test.ts
- packages/graph-core/src/__tests__/domain-persistence.test.ts

- packages/graph-core/src/__tests__/domain-types.test.ts
- packages/graph-core/src/__tests__/embedding-search.test.ts
- packages/graph-core/src/__tests__/fingerprint.test.ts
- packages/graph-core/src/__tests__/framework-registry.test.ts
- packages/graph-core/src/__tests__/ignore-filter.test.ts
- packages/graph-core/src/__tests__/ignore-generator.test.ts
- packages/graph-core/src/__tests__/language-lesson.test.ts
- packages/graph-core/src/__tests__/language-registry.test.ts
- packages/graph-core/src/__tests__/layer-detector.test.ts
- packages/graph-core/src/__tests__/normalize-graph.test.ts
- packages/graph-core/src/__tests__/parsers.test.ts
- packages/graph-core/src/__tests__/plugin-discovery.test.ts
- packages/graph-core/src/__tests__/plugin-registry.test.ts
- packages/graph-core/src/__tests__/schema.test.ts
- packages/graph-core/src/__tests__/search.test.ts
- packages/graph-core/src/__tests__/staleness.test.ts
- packages/graph-core/src/__tests__/tour-generator.test.ts
- packages/graph-core/src/analyzer/graph-builder.test.ts
- packages/graph-core/src/analyzer/graph-builder.ts
- packages/graph-core/src/analyzer/language-lesson.ts
- packages/graph-core/src/analyzer/layer-detector.ts
- packages/graph-core/src/analyzer/llm-analyzer.test.ts
- packages/graph-core/src/analyzer/llm-analyzer.ts
- packages/graph-core/src/analyzer/normalize-graph.ts
- packages/graph-core/src/analyzer/tour-generator.ts
- packages/graph-core/src/change-classifier.ts
- packages/graph-core/src/embedding-search.ts
- packages/graph-core/src/fingerprint.ts

Detected stacks (marker files)

Stack	Markers
node	package.json

Class / symbol index (heuristic)

File	Symbols / defs (regex)
cli/src/main.rs	ChangeKind, path_segments, path_depth, parent_dir, subtree_change_counts, try_build_gitignore_for_root, is_novadiff_docs_reserved_rel, filter_out_novadiff_docs_changes, walk_path_allowed, filter_changes_for_prefetch_gitignore, is_utf8_text_file, is_change_text_eligible, filter_out_non_text_changes, prefetch_summary_queue, empty_summary_evidence, compact_snippet, push_unique_string, line_range_label, detect_file_roles, build_changed_line_ranges, changed_lines_for_side, collect_touched_symbols, is_import_like_line, is_export_like_line, collect_changed_decl_lines, collect_cited_changed_lines, build_verification_hints, build_evidence_badges
electron/code-city-model.cjs	normalizeSide, dominantAuthor, ownershipForRange, selectBlamePaths, buildBlameMap, buildSymbolFileIndex, buildSideNodes, buildCodeCityModelPayload
electron/compare-runner.cjs	cliExecutableName, resolveRustCli, cleanEnv, runCompareEngine, parseEngineStdout, runCompareEngineAsync
electron/diff-excerpt.cjs	buildDiffExcerpt, buildDiffExcerptChunks
electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/file-summary-export.cjs	escapeHtml, escapeMermaidForHtml, injectMermaidFences, markdownToHtml, isReservedNovadiffRel, safeSummarySegments, resolveSummaryDirAbs, safeArtifactSegment, resolveSelectionBaseDirAbs, resolveSelectionDirAbs, buildFileSummaryDocument, buildSelectionSummaryDocument, exportFileSummaryArtifacts, readFileSummaryMarkdowns, exportSelectionSummaryArtifacts, readSelectionSummaryMarkdowns
electron/gh-install.cjs	resolveBrew, getInstallPlan, readGhVersion, getGhToolingStatus, runInstallCommand, installGh
electron/gh-path.cjs	homeDir, extraPathDirs, augmentPathForCli, ghCandidatePaths, resolveViaLoginShell, verifyGh, resolveGhExecutable, clearGhCache, isGhInstalled, ghInstallHint, ghNotFoundError
electron/git-blame.cjs	parseBlamePorcelain, blameFileAtRef, blameFileOwnership
electron/git-commit.cjs	getCommitDetail
electron/git-publish.cjs	buildFullCommitMessage, previewPublish, executePublish
electron/git-service.cjs	gitExecutable, runGit, tryRunGit, isGitRepo, parsePorcelainStatus, getRepoStatus, parseGithubSlugFromUrl, stageAll, stagePaths, unstagePaths, stageDistrict, commit, push, ensureWorktreeBase, removeDirSafe, checkoutRefWorktree, cleanupNovadiffWorktrees, preparePrCompareRoots, splitGitFormatLine, parseLogLines, listBranches, listCommitsForRef, resolveRefHash, detectGitTooling
electron/github-auth-flow.cjs	stripAnsi, cancelGithubDeviceAuth, startGithubDeviceAuth
electron/github-commit-context.cjs	parseJsonSafe, shortGhError, normalizeThread, getGithubCommitContext
electron/github-service.cjs	runGh, tryRunGh, isGhAvailable, getAuthStatus, listRepos, listPullRequests, viewPullRequest, createPullRequest, getUserProfile, listDetectedAccounts, slugFromRepoRoot
electron/graph/extract-structure.mjs	main, buildResult
electron/knowledge-graph-runner.cjs	resolveRepoRoot, resolveNodeExecutable, nodeChildEnv, runCommand, ensureGraphCoreBuilt, loadCore, isTextCandidate, walkProjectFiles, detectLanguage, fileCategory, resolveRelativeImport, buildBatchImportData, runExtractStructureBatch, mapStructuralAnalysis, heuristicComplexity, buildFileMeta, topSegment, buildLayersFromFiles, assignNodeLayers, buildDiffOverlay, getGitHash, buildKnowledgeGraph
electron/llm.cjs	normalizeBase, sendAccumulated, normalizeRepeatUnit, arraysEqual, trimRepeatedParagraphSuffix, trimRepeatedLineSuffix, sanitizeModelResponse, ollamaGenerationOptions, lmStudioGenerationOptions, ollamaNumPredict, streamOllamaChat, lmStudioMaxTokens, streamLmStudioChat, buildWorkspaceDocPrompt, buildCodebaseDocPrompt, buildCommitMessagePrompt, buildSelectionDocPrompt, bulletBlock, renderSummaryEvidenceBlock, buildExplainCodePrompt, buildGroundingContext, verifierWordLimit, buildGroundedRevisionPrompt, shouldRunVerifier, reviseGroundedMarkdown, buildPerFileSummarizePrompt, buildChunkExtractionPrompt, buildChunkSynthesisPrompt

File	Symbols / defs (regex)
electron/main.cjs	sendEngineProgress, getTargetWindow, windowStatePayload, emitWindowState, iconPath, buildMenu, createWindow, sendKnowledgeGraphProgress, sendWorkspaceHistoryProgress, sendGithubAuthProgress
electron/novadiff-docs-html.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, escapeRegex, countMatches, normalizeMalformedCodeFences, looksLikeRawDiffDump, normalizeFenceChunk, normalizeCompactListMarkers, injectMermaidFences, normalizeMarkdownDocument, stripTags, slugify, addHeadingAnchors, markdownToHtml, mermaidSection, navHtml, topDirectory, buildImpactHeatmapHtml, buildSearchIndex, buildEnhancementScript, wrapPage, bundleMetaLine, buildIndexHtml, buildNarrativeHtml, buildMetricsHtml
electron/novadiff-docs-pdf.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, renderSymbolsTable, renderImportEdgesTable, renderDetectedProjects, renderCrossFileCallEdges, mermaidBlock, buildMainPdfHtml, buildDiagramsPdfHtml, waitForPdfReady, htmlToPdfFile, generateNovadiffDocsPdf
electron/novadiff-docs-writer.cjs	bundleKeyOf, bundleBaseDir, bundleHeadline, readme, writeNovadiffDocsBundle
electron/prefetch-summaries.cjs	clearSummaryPrefetchCache, stopSummaryPrefetchWorker, getPrefetchedSummary, defaultPrefetchLimit, runSummaryPrefetchLoop, startSummaryPrefetchWorker
electron/repo-discovery.cjs	homeSearchRoots, readGitRemoteSlug, scanForGitRepos, discoverRepos, matchRepoForGithubRepo
electron/security-scan-runner.cjs	severityFromAdvisory, confidenceFromSeverity, hashId, relPathFromRoot, walkRepoForMarkers, detectEcosystems, runCommand, parseNpmAudit, parsePipAudit, parseCargoAudit, parseOsvScanner, runSecurityInsightScan
electron/workspace-files.cjs	listSnapshotFiles, readSnapshotTextFile
electron/workspace-history.cjs	removeDirSafe, isUsableSnapshotDir, listCommitsOldestFirst, snapshotCommit, ensureCommitSnapshot, indexWorkspaceHistory, refreshWorkspaceHistory, materializeCommitForCompare
electron/workspace-store.cjs	sessionPath, workspacesRoot, readJson, withSessionWriteLock, writeJsonAtomic, writeJson, loadSession, setLocalOnlyMode, saveSession, listWorkspaces, upsertWorkspace, setActiveWorkspace, setGitUser, getWorkspace, updateWorkspaceUiState, updateWorkspaceLiveRepo, createWorkspace
packages/graph-core/scripts/generate-alphabetical-catalog.mjs	normalizeKey, slugify, resolveWasmsKey, resolveExtensions, claimExtensions
packages/graph-core/src/__tests__/change-classifier.test.ts	makeAnalysis
packages/graph-core/src/__tests__/plugin-registry.test.ts	createMockPlugin
packages/graph-core/src/__tests__/staleness.test.ts	makeGraph
packages/graph-core/src/analyzer/graph-builder.ts	FileMeta, FileAnalysisMeta, NonCodeFileMeta, NonCodeFileAnalysisMeta, GraphBuilder
packages/graph-core/src/analyzer/language-lesson.ts	LanguageLessonResult, buildConceptPatterns, detectLanguageConcepts, getLanguageDisplayName, buildLanguageLessonPrompt, extractJson, parseLanguageLessonResponse
packages/graph-core/src/analyzer/layer-detector.ts	LLMLayerResponse, toLayerId, matchFileToLayer, detectLayers, buildLayerDetectionPrompt, parseLayerDetectionResponse, applyLLMLayers
packages/graph-core/src/analyzer/llm-analyzer.ts	LLMFileAnalysis, LLMProjectSummary, buildFileAnalysisPrompt, buildProjectSummaryPrompt, extractJson, parseFileAnalysisResponse, parseProjectSummaryResponse
packages/graph-core/src/analyzer/normalize-graph.ts	stripToValidPrefix, normalizeNodeId, normalizeComplexity, DroppedEdge, NormalizationStats, NormalizeBatchResult, inferTypeFromId, normalizeBatchOutput
packages/graph-core/src/analyzer/tour-generator.ts	buildTourGenerationPrompt, parseTourGenerationResponse, generateHeuristicTour
packages/graph-core/src/change-classifier.ts	UpdateDecision, classifyUpdate, detectDirectoryChanges, topDirectory, summarizeChanges
packages/graph-core/src/embedding-search.ts	SemanticSearchOptions, cosineSimilarity, SemanticSearchEngine
packages/graph-core/src/fingerprint.ts	FunctionFingerprint, ClassFingerprint, ImportFingerprint, FileFingerprint, FingerprintStore, FileChangeResult, ChangeAnalysis, contentHash,

File	Symbols / defs (regex)
	extractFileFingerprint, compareFingerprints, buildFingerprintStore, analyzeChanges
packages/graph-core/src/ignore-filter.ts	IgnoreFilter, createIgnoreFilter
packages/graph-core/src/ignore-generator.ts	parseGitIgnorePatterns, isCoveredByDefaults, generateStarterIgnoreFile
packages/graph-core/src/languages/catalog-builder.ts	catalogEntryToLanguageConfig, buildAlphabeticalLanguageConfigs, listAlphabeticalLanguages
packages/graph-core/src/languages/catalog-types.ts	LanguageCatalogEntry
packages/graph-core/src/languages/framework-registry.ts	FrameworkRegistry
packages/graph-core/src/languages/language-registry.ts	LanguageRegistry
packages/graph-core/src/languages/merge-configs.ts	mergeLanguageConfigs, languageConfigsForRegistry
packages/graph-core/src/persistence/index.ts	ensureDir, sanitiseFilePaths, saveGraph, loadGraph, saveMeta, loadMeta, saveFingerprints, loadFingerprints, saveConfig, loadConfig, saveDomainGraph, loadDomainGraph
packages/graph-core/src/plugins/discovery.ts	PluginEntry, PluginConfig, parsePluginConfig, serializePluginConfig
packages/graph-core/src/plugins/extractors/__tests__/cpp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/csharp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/go-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/java-extractor.test.ts	parse, Repository
packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	parse, helper, greet, noReturn, noop, multiline, Loggable, Repository, publicFunc, anotherFunc, myFunc, process, main, helper, findUser
packages/graph-core/src/plugins/extractors/__tests__/python-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/ruby-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/rust-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/base-extractor.ts	traverse, getStringValue, findChild, findChildren, hasChildOfType
packages/graph-core/src/plugins/extractors/cpp-extractor.ts	unwrapDeclaratorName, extractFuncDeclName, extractParams, extractReturnType, isStatic, CppExtractor
packages/graph-core/src/plugins/extractors/csharp-extractor.ts	extractParams, extractReturnType, hasModifier, extractUsingSource, lastComponent, extractInvocationName, CSharpExtractor
packages/graph-core/src/plugins/extractors/generic-extractor.ts	nodeName, lineRange, importSource, GenericExtractor
packages/graph-core/src/plugins/extractors/go-extractor.ts	extractParams, extractResultType, extractReceiverType, isExported, GoExtractor
packages/graph-core/src/plugins/extractors/java-extractor.ts	extractParams, extractReturnType, hasModifier, extractScopedIdentifierPath, lastComponent, JavaExtractor
packages/graph-core/src/plugins/extractors/php-extractor.ts	extractParams, extractReturnType, extractUseName, lastSegment, PhpExtractor
packages/graph-core/src/plugins/extractors/python-extractor.ts	extractParams, extractReturnType, unwrapDecorated, PythonExtractor

File	Symbols / defs (regex)
packages/graph-core/src/plugins/extractors/ruby-extractor.ts	extractParams, extractAttrProperties, getStringContent, RubyExtractor
packages/graph-core/src/plugins/extractors/rust-extractor.ts	extractParams, extractReturnType, isPublic, extractScopedPath, RustExtractor
packages/graph-core/src/plugins/extractors/types.ts	LanguageExtractor
packages/graph-core/src/plugins/extractors/typescript-extractor.ts	extractParams, extractReturnType, extractImportSpecifiers, TypeScriptExtractor

Module import edges (tabular)

Use together with the flowchart in the Diagrams PDF for a "call graph"-style overview at module boundaries.

From	To
electron/code-city-model.cjs	electron/git-blame.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs
electron/gh-install.cjs	electron/gh-path.cjs
electron/git-commit.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/github-service.cjs
electron/github-auth-flow.cjs	electron/gh-path.cjs
electron/github-commit-context.cjs	electron/github-service.cjs
electron/github-service.cjs	electron/git-service.cjs
electron/github-service.cjs	electron/gh-path.cjs
electron/github-service.cjs	electron/github-commit-context.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/main.cjs	electron/llm.cjs
electron/main.cjs	electron/prefetch-summaries.cjs
electron/main.cjs	electron/novadiff-docs-writer.cjs
electron/main.cjs	electron/novadiff-docs-pdf.cjs
electron/main.cjs	electron/novadiff-docs-html.cjs
electron/main.cjs	electron/file-summary-export.cjs
electron/main.cjs	electron/code-city-model.cjs
electron/main.cjs	electron/knowledge-graph-runner.cjs
electron/main.cjs	electron/security-scan-runner.cjs
electron/main.cjs	electron/git-service.cjs
electron/main.cjs	electron/github-service.cjs
electron/main.cjs	electron/git-commit.cjs
electron/main.cjs	electron/repo-discovery.cjs
electron/main.cjs	electron/git-publish.cjs
electron/main.cjs	electron/workspace-store.cjs
electron/main.cjs	electron/git-blame.cjs
electron/main.cjs	electron/workspace-files.cjs
electron/main.cjs	electron/workspace-history.cjs
electron/main.cjs	electron/github-auth-flow.cjs
electron/main.cjs	electron/gh-path.cjs
electron/main.cjs	electron/gh-install.cjs
electron/main.cjs	electron/export-snapshot.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/prefetch-summaries.cjs	electron/llm.cjs
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs
electron/prefetch-summaries.cjs	electron/file-summary-export.cjs
electron/prefetch-summaries.cjs	electron/git-service.cjs
electron/repo-discovery.cjs	src/app/types.ts
electron/security-scan-runner.cjs	electron/git-service.cjs
electron/workspace-history.cjs	electron/git-service.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
packages/graph-core/src/__tests__/ignore-filter.test.ts	packages/graph-core/src/ignore-filter.ts
packages/graph-core/src/__tests__/ignore-generator.test.ts	packages/graph-core/src/ignore-generator.ts

Cross-file call edges (tabular)

Edges appear only when a call site name uniquely resolves to a single definition path (very conservative).

From file	To file	Callee
cli/src/main.rs	electron/git-service.cjs	push
electron/code-city-model.cjs	electron/git-blame.cjs	blameFileOwnership
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs	normalizeMarkdownDocument
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs	htmlToPdfFile
electron/gh-install.cjs	electron/gh-path.cjs	augmentPathForCli
electron/gh-install.cjs	electron/gh-path.cjs	isGhInstalled
electron/gh-install.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/gh-install.cjs	electron/gh-path.cjs	clearGhCache
electron/git-commit.cjs	electron/git-service.cjs	runGit
electron/git-commit.cjs	electron/git-service.cjs	tryRunGit
electron/git-publish.cjs	electron/git-service.cjs	isGitRepo
electron/git-publish.cjs	electron/git-service.cjs	getRepoStatus
electron/git-publish.cjs	electron/github-service.cjs	getAuthStatus
electron/git-publish.cjs	electron/github-service.cjs	slugFromRepoRoot
electron/git-publish.cjs	electron/git-service.cjs	stagePaths
electron/git-publish.cjs	electron/git-service.cjs	stageAll
electron/git-publish.cjs	electron/git-service.cjs	commit
electron/git-publish.cjs	electron/git-service.cjs	push
electron/git-publish.cjs	electron/github-service.cjs	createPullRequest
electron/github-auth-flow.cjs	electron/gh-path.cjs	isGhInstalled
electron/github-auth-flow.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-auth-flow.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-auth-flow.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-commit-context.cjs	electron/github-service.cjs	getAuthStatus
electron/github-commit-context.cjs	electron/github-service.cjs	tryRunGh
electron/github-service.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-service.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-service.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-service.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/graph/extract-structure.mjs	packages/graph-core/src/plugins/parsers/index.ts	registerAllParsers
electron/knowledge-graph-runner.cjs	packages/graph-core/src/analyzer/graph-builder.ts	GraphBuilder
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveGraph
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveMeta
electron/llm.cjs	packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	process
electron/main.cjs	electron/gh-path.cjs	augmentPathForCli
electron/main.cjs	electron/compare-runner.cjs	runCompareEngineAsync

From file	To file	Callee
electron/main.cjs	electron/compare-runner.cjs	runCompareEngine
electron/main.cjs	electron/llm.cjs	summarizeChange
electron/main.cjs	electron/llm.cjs	summarizeChangeStream
electron/main.cjs	electron/llm.cjs	probeProvider
electron/main.cjs	electron/prefetch-summaries.cjs	stopSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	clearSummaryPrefetchCache
electron/main.cjs	electron/prefetch-summaries.cjs	startSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	getPrefetchedSummary
electron/main.cjs	electron/file-summary-export.cjs	exportFileSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readFileSummaryMarkdowns
electron/main.cjs	electron/file-summary-export.cjs	exportSelectionSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readSelectionSummaryMarkdowns
electron/main.cjs	electron/security-scan-runner.cjs	runSecurityInsightScan
electron/main.cjs	electron/code-city-model.cjs	buildCodeCityModelPayload
electron/main.cjs	electron/novadiff-docs-writer.cjs	writeNovadiffDocsBundle
electron/main.cjs	electron/novadiff-docs-html.cjs	writeNovadiffDocsHtml
electron/main.cjs	electron/novadiff-docs-pdf.cjs	generateNovadiffDocsPdf
electron/main.cjs	electron/novadiff-docs-html.cjs	refreshNovadiffDocsHtmlFromDisk
electron/main.cjs	electron/knowledge-graph-runner.cjs	buildKnowledgeGraph
electron/main.cjs	electron/git-service.cjs	detectGitTooling
electron/main.cjs	electron/github-service.cjs	getAuthStatus
electron/main.cjs	electron/git-service.cjs	getRepoStatus
electron/main.cjs	electron/repo-discovery.cjs	discoverRepos
electron/main.cjs	electron/repo-discovery.cjs	matchRepoForGithubRepo
electron/main.cjs	electron/github-service.cjs	listRepos
electron/main.cjs	electron/github-service.cjs	listPullRequests
electron/main.cjs	electron/git-service.cjs	isGitRepo
electron/main.cjs	electron/git-service.cjs	preparePrCompareRoots
electron/main.cjs	electron/github-service.cjs	viewPullRequest
electron/main.cjs	electron/git-publish.cjs	previewPublish
electron/main.cjs	electron/git-publish.cjs	executePublish
electron/main.cjs	electron/git-service.cjs	stagePaths
electron/main.cjs	electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/main.cjs	electron/git-service.cjs	stageDistrict
electron/main.cjs	electron/workspace-store.cjs	loadSession
electron/main.cjs	electron/github-service.cjs	listDetectedAccounts
electron/main.cjs	electron/github-service.cjs	getUserProfile
electron/main.cjs	electron/gh-install.cjs	getGhToolingStatus
electron/main.cjs	electron/gh-install.cjs	installGh
electron/main.cjs	electron/workspace-store.cjs	setLocalOnlyMode
electron/main.cjs	electron/github-auth-flow.cjs	startGithubDeviceAuth
electron/main.cjs	electron/github-auth-flow.cjs	cancelGithubDeviceAuth
electron/main.cjs	electron/workspace-store.cjs	setGitUser
electron/main.cjs	electron/workspace-store.cjs	workspacesRoot
electron/main.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/main.cjs	electron/workspace-store.cjs	createWorkspace
electron/main.cjs	electron/workspace-history.cjs	indexWorkspaceHistory
electron/main.cjs	electron/workspace-store.cjs	getWorkspace
electron/main.cjs	electron/workspace-store.cjs	setActiveWorkspace
electron/main.cjs	electron/workspace-store.cjs	listWorkspaces
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceLiveRepo
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceUiState
electron/main.cjs	electron/git-blame.cjs	blameFileAtRef
electron/main.cjs	electron/git-commit.cjs	getCommitDetail

Codebase outline (JSON excerpt)

```
{
  "by_extension": [
    {
      "count": 226,
      "ext": ".ts"
    },
    {
      "count": 85,
      "ext": ".tsx"
    },
    {
      "count": 58,
      "ext": ".pak"
    },
    {
      "count": 28,
      "ext": ".cjs"
    },
    {
      "count": 20,
      "ext": "(no ext)"
    },
    {
      "count": 14,
      "ext": ".json"
    },
    {
      "count": 9,
      "ext": ".plist"
    },
    {
      "count": 6,
      "ext": ".css"
    },
    {
      "count": 4,
      "ext": ".dylib"
    },
    {
      "count": 4,
      "ext": ".mjs"
    },
    {
      "count": 4,
      "ext": ".svg"
    },
    {
      "count": 3,
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    {
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      "ext": ".yaml"
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    {
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    {
      "count": 2,
      "ext": ".icns"
    },
    {
      "count": 1,
      "ext": ".asar"
    },
    {
      "count": 1,
      "ext": ".bin"
    },
    {
      "count": 1,
```

```

    "ext": ".dat"
  },
  {
    "count": 1,
    "ext": ".dmg"
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    "count": 1,
    "ext": ".ico"
  },
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  },
  {
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  {
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  },
  {
    "count": 1,
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  {
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  },
  {
    "count": 1,
    "ext": ".toml"
  },
  {
    "count": 1,
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  },
  {
    "count": 1,
    "ext": ".yaml"
  },
  {
    "count": 1,
    "ext": ".zip"
  }
],
"class_diagram_mermaid": "classDiagram\n class CSharpExtractor\n class ChangeAnalysis\n class ChangeKind\n class ClassFingerprint\n class CppExtractor\n class DockerfileParser\n class DroppedEdge\n class EnvParser\n class FileAnalysisMeta\n class FileChangeResult\n class FileFingerprint\n class FileMeta\n class FingerprintStore\n class FrameworkRegistry\n class FunctionFingerprint\n class GenericExtractor\n class GoExtractor\n class GraphBuilder\n class GraphQLParser\n class IgnoreFilter\n class ImportFingerprint\n class JSONConfigParser\n class JavaExtractor\n class LLMFileAnalysis",
"cross_file_call_edges": [
  {
    "from_file": "cli/src/main.rs",
    "to_file": "electron/git-service.cjs",
    "via": "push"
  },
  {
    "from_file": "electron/code-city-model.cjs",
    "to_file": "electron/git-blame.cjs",
    "via": "blameFileOwnership"
  },
  {
    "from_file": "electron/file-summary-export.cjs",
    "to_file": "electron/novadiff-docs-html.cjs",
    "via": "normalizeMarkdownDocument"
  },
  {
    "from_file": "electron/file-summary-export.cjs",
    "to_file": "electron/novadiff-docs-pdf.cjs",
    "via": "htmlToPdfFile"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "augmentPathForCli"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",

```

```

    "via": "isGhInstalled"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "resolveGhExecutable"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "clearGhCache"
  },
  {
    "from_file": "electron/git-commit.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "runGit"
  },
  {
    "from_file": "electron/git-commit.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "tryRunGit"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "isGitRepo"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "getRepoStatus"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/github-service.cjs",
    "via": "getAuthStatus"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/github-service.cjs",
    "via": "slugFromRepoRoot"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "stagePaths"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "stageAll"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "commit"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "push"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/github-service.cjs",
    "via": "createPullRequest"
  },
  {
    "from_file": "electron/github-auth-flow.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "isGhInstalled"
  },
  {
    "from_file": "electron/github-auth-flow.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "ghNotFoundError"
  },
  {
    "from_file": "electron/github-auth-flow.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "resolveGhExecutable"
  },

```

```

{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-commit-context.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "getAuthStatus"
},
{
  "from_file": "electron/github-commit-context.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "tryRunGh"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "ghNotFoundError"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "parseGithubSlugFromUrl"
},
{
  "from_file": "electron/graph/extract-structure.mjs",
  "to_file": "packages/graph-core/src/plugins/parsers/index.ts",
  "via": "registerAllParsers"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/analyzer/graph-builder.ts",
  "via": "GraphBuilder"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/persistence/index.ts",
  "via": "saveGraph"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/persistence/index.ts",
  "via": "saveMeta"
},
{
  "from_file": "electron/llm.cjs",
  "to_file": "packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts",
  "via": "process"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/compare-runner.cjs",
  "via": "runCompareEngineAsync"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/compare-runner.cjs",
  "via": "runCompareEngine"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/llm.cjs",
  "via": "summarizeChange"
},
{
  "from_file": "electron/main.cjs",

```

```
"to_file": "electron/llm.cjs",
"via": "summarizeChangeStream"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/llm.cjs",
  "via": "probeProvider"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/prefetch-summaries.cjs",
  "via": "stopSummaryPrefetchWorker"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/prefetch-summaries.cjs",
  "via": "clearSummaryPrefetchCache"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/prefetch-summaries.cjs",
  "via": "startSummaryPrefetchWorker"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/prefetch-summaries.cjs",
  "via": "getPrefetchedSummary"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "exportFileSummaryArtifacts"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "readFileSummaryMarkdowns"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "exportSelectionSummaryArtifacts"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "readSelectionSummaryMarkdowns"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/security-scan-runner.cjs",
  "via": "runSecurityInsightScan"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/code-city-model.cjs",
  "via": "buildCodeCityModelPayload"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-writer.cjs",
  "via": "writeNovadiffDocsBundle"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-html.cjs",
  "via": "writeNovadiffDocsHtml"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-pdf.cjs",
  "via": "generateNovadiffDocsPdf"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-html.cjs",
  "via": "refreshNovadiffDocsHtmlFromDisk"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/knowledge-graph-runner.cjs",
  "via": "buildKnowledgeGraph"
}
```

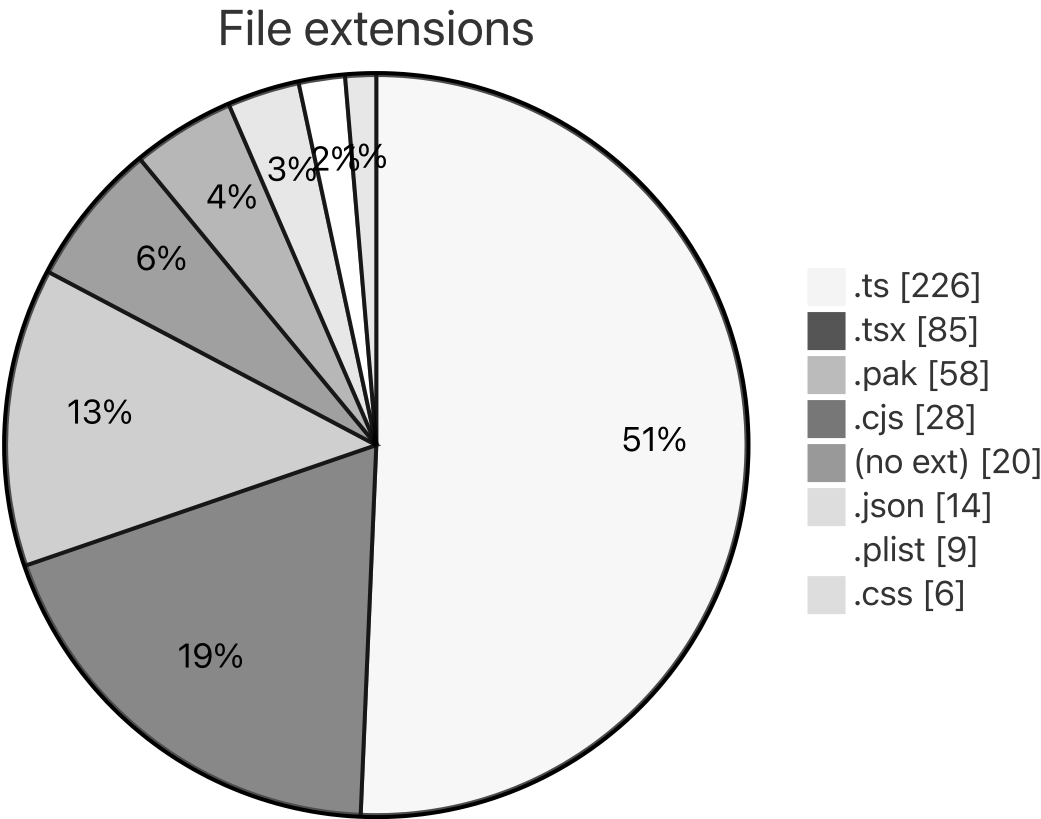
```

},
{
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  "to_file": "electron/git-service.cjs",
  "via": "detectGitTooling"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "getAuthStatus"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "getRepoStatus"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/repo-discovery.cjs",
  "via": "discoverRepos"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/repo-discovery.cjs",
  "via": "matchRepoForGithubRepo"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "listRepos"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "listPullRequests"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "isGitRepo"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "preparePrCompareRoots"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "viewPullRequest"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-publish.cjs",
  "via": "previewPublish"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-publish.cjs",
  "via": "executePublish"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "stagePaths"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/export-snapshot.cjs",
  "via": "exportProjectSnapshotZip"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "stageDistrict"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/workspace-store
... truncated ...

```

Diagrams (Mermaid)

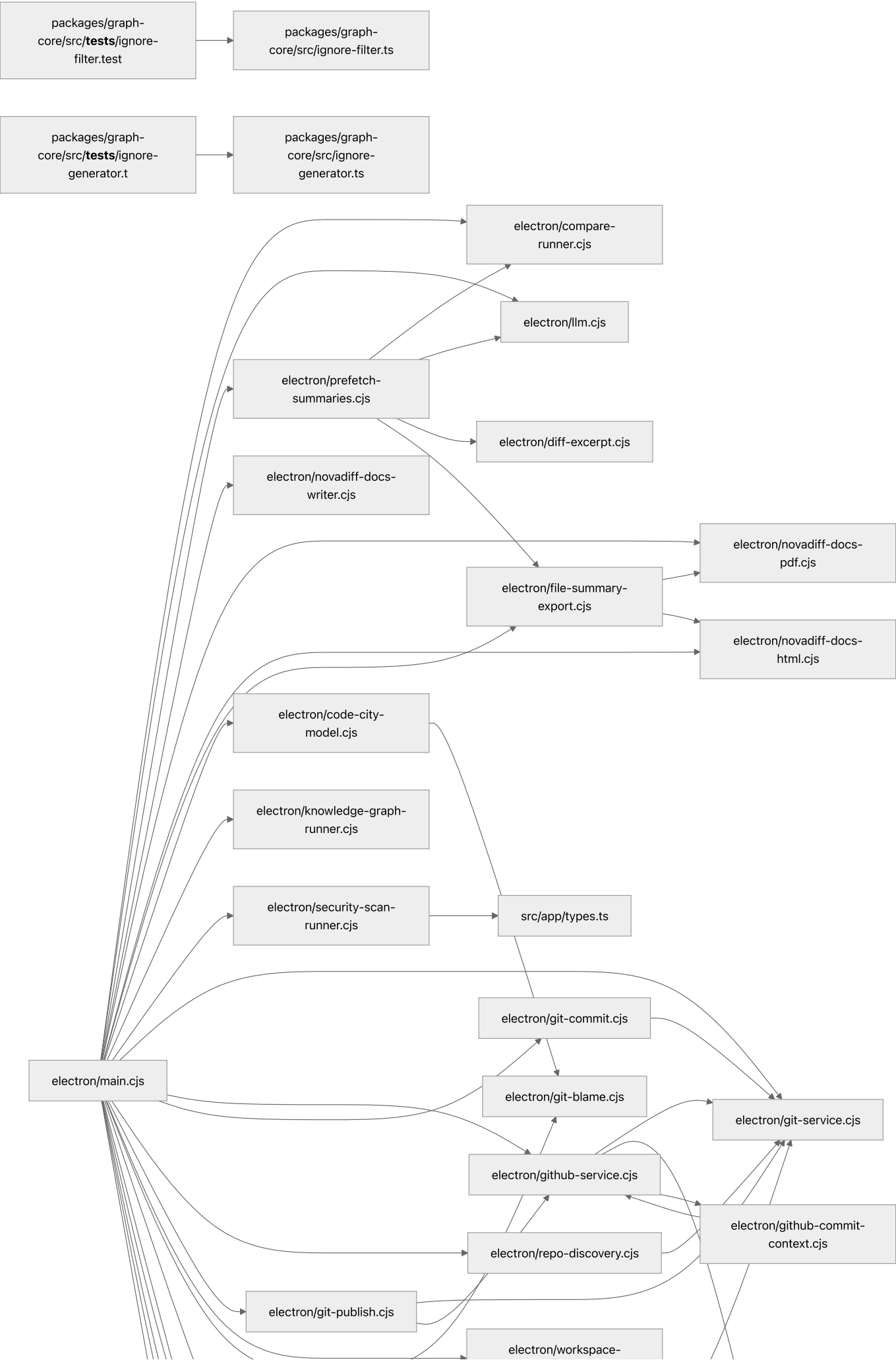
File extension mix

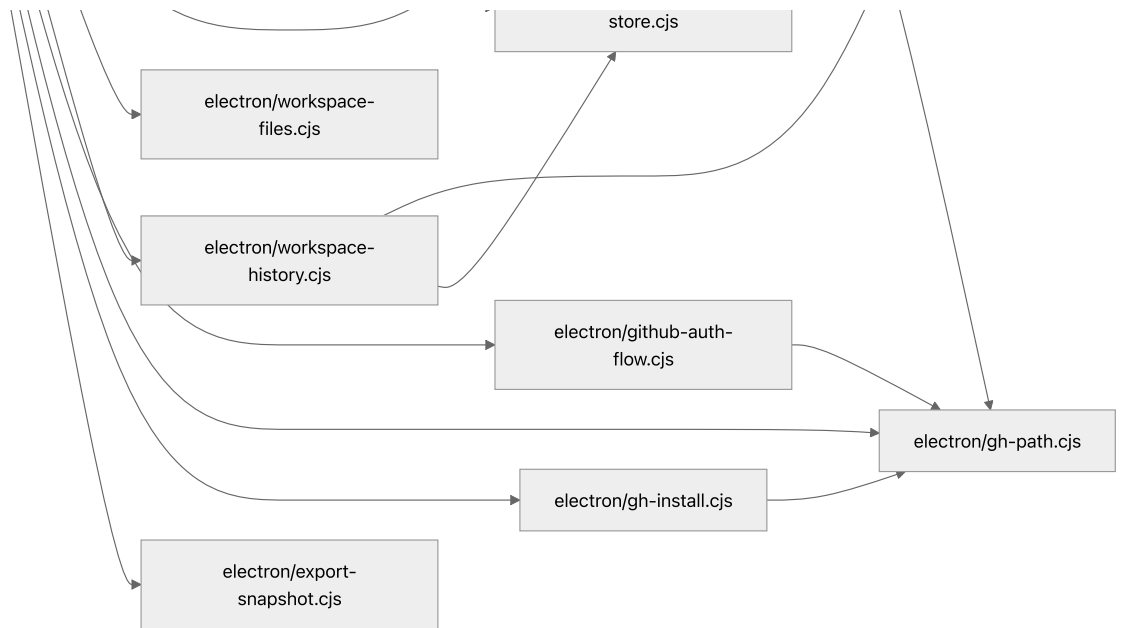


Directory depth



Module import graph (relative imports)





Cross-file calls (heuristic)

